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Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

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Abstract. Intrusion Detection Systems (IDS) are essential for securing IoT networks; however, most deep learning–based solutions are computationally intensive and unsuitable for resource-constrained edge devices. This paper proposes TSANet, a lightweight Temporal–Spatial Attention Network for real-time intrusion detection on low-power hardware such as Raspberry Pi 5. The model combines a 1D CNN for spatial feature extraction, an LSTM for temporal modeling, and attention mechanisms to capture important traffic patterns. To enable efficient deployment, pruning and quantization-aware training are applied to reduce model size and inference latency.TSANet is evaluated on the CICIDS2017 dataset and validated through real-time traffic monitoring on Raspberry Pi.The optimized model achieves 98.3% offline accuracy and 96.8% real-time accuracy,with low latency of 8–12 ms per flow. Unlike conventional IDS approaches focused on offline performance, this work emphasizes lightweight design and real-time edge deployment for practical IoT security.

Keywords: Intrusion Detection System · IoT Security · Edge Computing · Raspberry Pi · Deep Learning.

1 Introduction

This work aligns with UN Sustainable Development Goal 9, emphasizing secure digital infrastructure. The rapid growth of IoT devices in smart environments has increased exposure to cyber threats such as DoS, DDoS, and malware, necessitating efficient intrusion detection mechanisms. Intrusion Detection Systems (IDS) are widely used to monitor network traffic and detect malicious activities. Traditional IDS approaches rely on signature-based or rule-based methods, while recent deep learning–based IDS models have shown promising performance in identifying complex attack patterns. However, most of these models are computationally intensive and are typically deployed on high-end servers or cloud environments.

Deploying such models directly on IoT edge devices is challenging due to their limited CPU capacity, memory constraints, and energy efficiency requirements. In addition, real-time intrusion detection requires continuous packet capture, feature extraction, and low-latency inference, which further increases the computational burden. Offloading data to cloud servers also introduces latency and potential privacy risks. These limitations highlight the need for lightweight and efficient IDS solutions that can operate directly on edge devices.

To address these challenges, we propose TSANet (Temporal–Spatial Attention Network), a lightweight deep learning model designed for real-time intrusion detection on resource-constrained IoT devices. The proposed model performs end-to-end processing, including traffic analysis and inference, directly on the device. It is implemented on a Raspberry Pi 5 and evaluated using both benchmark datasets and live network traffic. Unlike existing CNN–LSTM-based IDS models that primarily focus on offline accuracy, the proposed approach emphasizes lightweight design, model optimization, and real-time deployment on edge devices. Experimental results demonstrate that effective intrusion detection can be achieved on IoT edge devices with high accuracy and low latency. **The main contributions of this work are summarized as follows:**

- A lightweight Temporal–Spatial Attention Network (TSANet) for intrusion detection in IoT environments.
- Integration of spatial (CNN-based) and temporal (LSTM-based) attention mechanisms for improved feature representation.
- Design of a real-time intrusion detection pipeline capable of processing live network traffic.
- Deployment and validation of the proposed model on a resource-constrained Raspberry Pi 5 device.
- Model optimization using pruning and quantization techniques to reduce computational complexity and inference latency.

2 Related Work

Intrusion detection in Internet of Things (IoT) networks has gained significant attention due to the rapid growth of connected devices in smart homes, cities, and industrial systems. Existing research spans traditional rule-based methods, deep learning–based intrusion detection systems (IDS), and lightweight approaches designed for resource-constrained environments. A summary of key contributions is presented in Table 1.

2.1 Traditional and Early IDS Approaches

Early intrusion detection approaches relied on machine learning and rule-based techniques to identify malicious network activity. While these methods laid the foundation for IDS, they do not scale effectively in dynamic and heterogeneous IoT environments. The importance of secure digital infrastructure is also emphasized in the United Nations Sustainable Development Goal 9 [1].

2.2 Deep Learning-Intrusion Detection Systems

Traditional techniques are insufficient for IoT environments; hence, deep learning approaches have been widely adopted. Convolutional Neural Networks (CNNs) are commonly used for spatial feature extraction in network traffic analysis [14]. Idrissi et al. [2] and Saba et al. [3] demonstrated improved detection using deep learning–based IDS models.

Hybrid architectures combining CNN and LSTM models have also been explored. Gueriani et al. [4] and Otoum and Nayak [5] leveraged spatial and temporal learning for improved detection. Survey studies further highlight the effectiveness of deep learning in IoT IDS while identifying challenges related to model complexity and deployment feasibility [6, 7].

2.3 Lightweight and Edge-Based IDS Solutions

Recent research focuses on lightweight and edge-deployable IDS models that reduce computational complexity while maintaining high detection accuracy. Kayode et al. [8] proposed a TSANet-based model for DoS detection; however, their work does not address real-time deployment challenges on resource-constrained devices. Racherla et al. [9] demonstrated real-time IDS feasibility in constrained environments.

Several works explore Raspberry Pi–based IDS deployment. Dahiya et al. [10], Olakanmi and Olatunji [11], Adrian et al. [12], and Gupta et al. [13] highlight edge-based detection capabilities. Despite these advances, most approaches focus on offline accuracy, with limited attention to real-time constraints such as latency, computational overhead, and continuous traffic processing on low-power edge devices.

Table 1: Summary of IoT Intrusion Detection Systems

Author	**Methodology**	**Dataset**	**Feature Extraction**	**Results**
Idrissi et al. (2022) [2]	Optimized deep learning HIDS for IoT edge	IoT system logs	Host-level statistical features	Improved accuracy with reduced overhead on edge devices
Saba et al. (2022) [3]	Anomaly-based deep learning IDS	IoT traffic	Statistical flow features	High detection rate for IoT attacks
Gueriani et al. (2023) [4]	Hybrid CNN–LSTM IDS	Simulated IoT traffic	Spatial and temporal flow features	Enhanced detection accuracy over single models

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Author	Methodology	Dataset	Feature Extraction	Results
Otoum and Nayak (2022) [5]	Deep learning–based IDS (DL-IDS)	IoT network traffic	Sequential traffic features	Effective detection of complex attacks
Kayode et al. (2025) [8]	Temporal–Spatial Attention Network (TSANet)	Network traffic traces	Attention-weighted flow features	High DoS detection accuracy using attention mechanisms
Racherla et al. (2024) [9]	Real-time deep IDS for IoT	IoT network traffic	Flow-based statistical features	Low-latency detection on IoT nodes
Dahiya et al. (2024) [10]	Raspberry Pi–based IDS	IoT traffic	Basic flow statistics	Feasible edge deployment with limited evaluation
Olakanmi and Olatunji (2024) [11]	Optimized ensemble deep learning IDS	IoT traffic	Reduced feature set	Improved accuracy with lower resource usage
Adrian et al. (2025) [12]	Adaptive ML-based IDS	Smart home traffic	Network flow features	Adaptive detection for smart home security
Gupta et al. (2022) [13]	Deep learning IDS for 5G-IoT	IoT over 5G networks	Traffic flow features	Low-latency detection in 5G-enabled IoT
Tsimenidis et al. (2022) [6]	Survey on DL-based IoT IDS	Multiple benchmarks	Comparative feature analysis	Identified challenges in DL IDS deployment
Morshedi et al. (2024) [7]	Deep learning IDS for IoT security	IoT datasets	Flow-level features	High accuracy but limited edge evaluation
Shone et al. (2018) [14]	Deep autoencoder-based IDS	NSL-KDD	Learned latent representa-tions	Improved detection of unseen attacks

2.4 Research Gap

There are many intrusion detection methods for IoT networks, but most of them are based on heavy deep learning models that are suitable for cloud or server-based systems. These models are not necessarily capable of real-time processing

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on edge devices. Most of the work done has shown high accuracy in offline scenarios but have not been thoroughly tested for latency, resource consumption, and robustness in real-world settings. Moreover, some of the techniques are only based on spatial feature learning or temporal feature learning and not both. There have been some efforts on model compression and federated learning.

3 Proposed Methodology

This section discusses the proposed lightweight intrusion detection system that utilizes the Temporal-Spatial Attention Network (TSANet). Its main contribution is a real-time-based deep learning intrusion detection system that operates solely on a Raspberry Pi-based device, which analyzes the incoming network traffic in real time. Unlike the traditional intrusion detection system architecture that relies on a centralized processing system, this system operates solely on the edge device.

3.1 System Overview

The overall workflow of the proposed system is shown in Figure 3. Real-time network packets are captured, converted into bidirectional flows, represented as numerical features, and passed to the trained TSANet model deployed on the Raspberry Pi for classification.

3.2 Data Collection and Feature Engineering

Data collection is performed in two phases. The CICIDS2017 dataset is used for offline training, while real-time traffic is captured on a Raspberry Pi configured as a Wi-Fi access point.

Each network connection is represented using 32 flow-based features extracted from bidirectional traffic, including packet statistics, flow duration, and byte-level information. The feature subset is selected based on domain knowledge to capture essential traffic characteristics.

A reduced feature set is used instead of the full dataset to minimize memory usage, computational overhead, and inference latency, making it suitable for edge deployment. Flow-based features provide a compact and efficient representation for real-time analysis.

All features are normalized using Min-Max scaling to ensure stable model performance. Formal feature importance ranking is not performed, and selection is guided by domain knowledge and deployment constraints.

3.3 Data Preprocessing

The preprocessing pipeline includes: (1) cleaning of the dataset by removing invalid or missing entries, (2) conversion of raw packet data into bidirectional flow-based representations, (3) selection of 32 relevant numerical features, (4)

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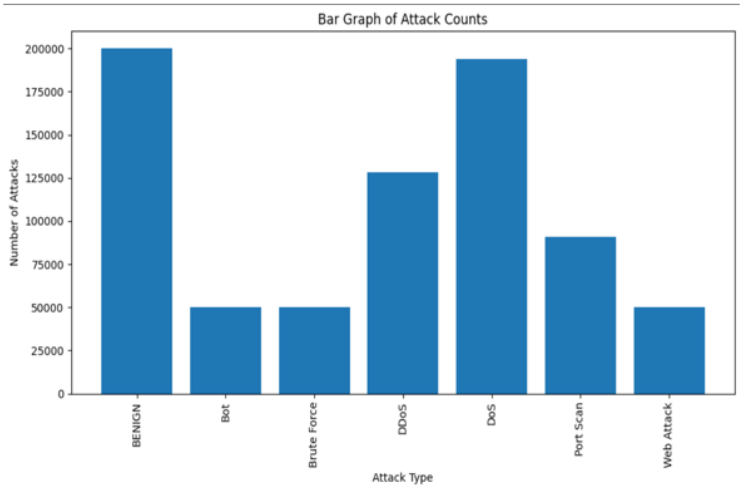

Fig. 1. Fig. 1. Distribution of benign and attack samples across seven traffic categories in the CICIDS2017 dataset, illustrating significant class imbalance particularly for rare attack types such as Heartbleed.

normalization using Min-Max scaling to ensure stable model training, and (5) label encoding of attack categories for supervised classification. The same pipeline is consistently applied during both offline training and real-time inference on Raspberry Pi to maintain model reliability..

3.4 TSANet Architecture

At the core of the proposed system is the TSANet model, shown in Figure 2. The architecture is designed to capture spatial and temporal characteristics of network traffic in a structured manner.

First, the input feature vector is processed using a one-dimensional Convolutional Neural Network (CNN) with 32 filters to extract spatial relationships between traffic features. This is followed by a 1×1 convolution layer that generates spatial attention weights, which are applied through element-wise multiplication to emphasize the most informative features.

Next, temporal dependencies in the traffic flow are modeled using an LSTM layer with 32 hidden units. A dense layer with sigmoid activation computes temporal attention weights, allowing the model to focus on the most important time steps in the sequence. Batch normalization is used to enhance training stability.

The outputs from both spatial and temporal attention branches are fused by concatenation and passed through an LSTM layer with 64 units to jointly learn spatio-temporal patterns. Finally, fully connected layers map the learned features to the output space, producing class probabilities for benign and malicious traffic.

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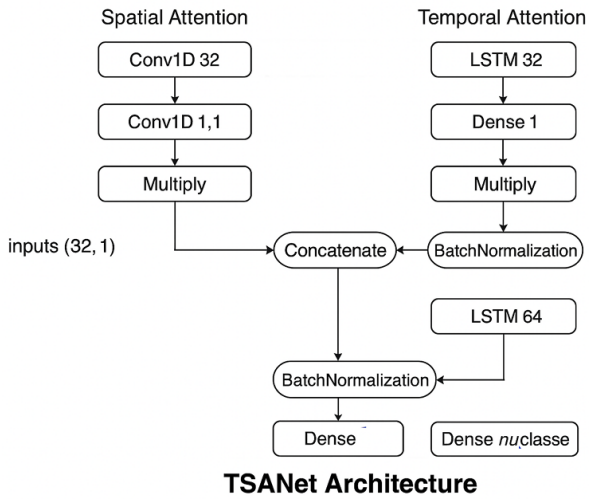

Fig. 2. TSANet architecture illustrating the spatial attention branch (1D CNN-based), temporal attention branch (LSTM-based), feature fusion via concatenation, and final classification layers.

3.5 Model Training and Optimization

TSANet is trained using categorical cross-entropy loss:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) \tag{1}$$

where $\hat{y}_c$ is the predicted probability for class c.

The model is trained using the Adam optimizer (learning rate 0.001), with a batch size of 64 for 30 epochs. Training is performed on a standard system (Intel i5/i7, 8–16 GB RAM) and takes approximately 25–40 minutes. Inference is executed on a Raspberry Pi 5 to validate real-time deployment. To enable efficient edge deployment, pruning and quantization-aware training are applied to reduce model size and inference latency. The optimized model is exported using TorchScript.

3.6 Edge Deployment on Raspberry Pi

The improved TSANet model is implemented on a Raspberry Pi 5 to function as a real-time network gateway. On the hardware side, Scapy is used for packet capture, followed by flow reconstruction and feature extraction on the same hardware. Model inference is conducted on the same hardware using the TorchScript model.

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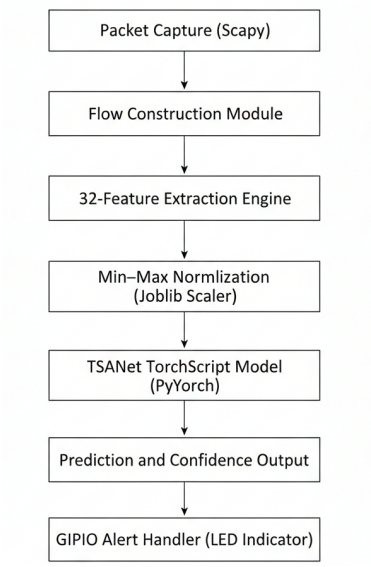

Fig. 3. End-to-end system workflow for real-time traffic capture, feature extraction, and edge inference

Figure 3 illustrates the complete runtime pipeline, including packet capture, feature processing, model inference, and alert generation. Upon detecting malicious activity, the system outputs prediction confidence and triggers a GPIO-based LED indicator for immediate physical alerting. This fully autonomous design eliminates reliance on external servers and preserves data privacy.

Overall, the proposed methodology demonstrates that effective intrusion detection can be achieved directly at the IoT edge by integrating attention-driven deep learning with lightweight deployment strategies.

4 Results

In this section, we will present the experimental results of the TSANet-based intrusion detection system on the standard benchmark datasets.

4.1 Offline Evaluation on CICIDS2017

We conducted an offline evaluation of the TSANet model using the CICIDS2017 dataset based on standard metrics: accuracy, precision, recall, and F1-score.

Table 2 compares TSANet with baseline models. The proposed model achieves the highest accuracy of 98.3%, outperforming both traditional machine learning and deep learning approaches.

Recent lightweight intrusion detection models focus on reducing computational complexity through feature reduction and model compression. In contrast,

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TSANet integrates spatial and temporal attention mechanisms with a compact feature set and further employs pruning and quantization, enabling efficient real-time deployment while maintaining high detection performance. The improved performance is attributed to the model's ability to capture both spatial feature relationships and temporal dependencies, allowing it to effectively learn complex attack patterns.

Table 2. Table 2. Comparison of TSANet with baseline models on CICIDS2017 dataset across accuracy, precision, recall, and F1-score (all values in %).

Model	Accuracy	Precision	Recall	F1-score
Random Forest	94.2	93.7	92.5	92.8
CNN	95.6	95.1	94.8	94.9
LSTM	96.2	96.0	95.7	95.8
CNN–LSTM	97.0	96.9	96.4	96.6
Proposed TSANet	**98.3**	**98.6**	**97.9**	**98.1**

4.2 Effect of Model Pruning

To optimize TSANet for edge deployment, structured pruning was applied. Figure 4 shows the relationship between pruning rate and classification accuracy. The results indicate that accuracy remains stable up to moderate pruning levels, demonstrating that TSANet retains robustness even after model compression.

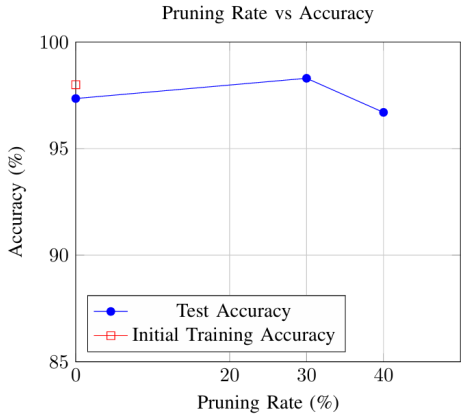

Fig. 4. Impact of pruning rate on classification accuracy demonstrating that TSANet retains stable performance up to moderate pruning levels before accuracy degrades.

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4.3 Analysis of Confusion Matrix

The confusion matrix for the optimized TSANet model on the CICIDS2017 dataset is shown in Table 3. The model demonstrates high detection accuracy for dominant attack classes such as DoS and DDoS.

However, the CICIDS2017 dataset exhibits class imbalance, where certain attack types such as Heartbleed have significantly fewer samples. This imbalance limits the model's ability to learn robust representations for minority classes, resulting in comparatively lower detection performance.

Although explicit imbalance handling techniques were not applied in this work, approaches such as SMOTE and weighted loss functions can improve sensitivity toward underrepresented classes and are considered for future work.

Furthermore, the evaluation is limited to a single dataset. Future work will include validation on additional IoT datasets such as UNSW-NB15 and IoT-23 to assess generalization and robustness.

Table 3. Presents the Classwise Prediction of the optimized TSAnet model (%) showing high performance on dominant classes and relatively lower detection for rare classes such as Heartbleed due to dataset imbalance.

Ground / Model OP	Benign	Bot	BF	DDoS	DoS	Heartbleed	Infiltration
Benign	98.1	0.2	0.3	0.1	0.8	0.0	0.5
Bot	0.4	97.6	1.2	0.0	0.3	0.0	0.5
Brute Force	0.6	0.8	96.4	0.0	1.7	0.0	0.5
DDoS	0.0	0.0	0.0	98.9	1.1	0.0	0.0
DoS	0.4	0.1	0.3	1.5	97.3	0.1	0.3
Heartbleed	0.0	0.0	0.0	0.0	2.5	95.8	1.7
Infiltration	1.3	0.4	0.0	0.0	0.7	0.2	97.4

4.4 Real-Time Deployment and Live Traffic Analysis

The optimized TSANet model is deployed on a Raspberry Pi 5 configured as a wireless access point (AP). The experimental setup comprised a Raspberry Pi 5 configured as a wireless access point, with three client devices (two laptops and one smartphone) connected to generate realistic mixed traffic including web browsing, video streaming, messaging, and background system processes, captured continuously over a 30-minute window.

The network traffic includes common user activities such as web browsing, video streaming, messaging, background updates, and system processes. All packets are captured using Scapy and converted into bidirectional flows for feature extraction. The same preprocessing pipeline used during offline training is applied before passing the flows to the deployed TSANet model for real-time

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```
===== ALERT =====
Flow: ('163.70.144.8', 443) → ('192.168.50.100', 45190)
Prediction: BENIGN (conf=0.5827)
=================

===== ALERT =====
Flow: ('192.168.50.140', 8080) → ('192.168.50.33', 41076)
Prediction: Heartbleed (conf=0.7354)
=================

===== ALERT =====
Flow: ('192.168.50.140', 52550) → ('34.107.221.82', 80)
Prediction: BENIGN (conf=0.5482)
================-

===== ALERT =====
Flow: ('192.168.50.140', 8080) → ('192.168.50.33', 41076)
Prediction: DDoS (conf=0.7714)
=================
```

Fig. 5. Real-time alerts generated by the deployed TSANet model, showing classification of network flows with associated confidence scores during live traffic monitoring on Raspberry Pi.

classification.A representation of real-time alerts generated during live traffic analysis is shown in Figure 5.

Each alert corresponds to a network flow and includes the predicted class along with a confidence score. Benign traffic generated by normal user activity is correctly classified, while malicious flows trigger high-confidence alerts, demonstrating the effectiveness of real-time edge-based intrusion detection.

5 Conclusion and Future Work

This paper presents TSANet, a lightweight Temporal-Spatial Attention Network designed for real-time intrusion detection in IoT environments. The proposed model integrates CNN-based spatial feature extraction, LSTM-based temporal modeling, and attention mechanisms to effectively capture complex network traffic patterns. To enable deployment on resource-constrained devices, the model is optimized using pruning and quantization techniques and implemented on a Raspberry Pi 5 for real-time edge-based intrusion detection. Experimental results on the CICIDS2017 dataset demonstrate high detection accuracy (98.3%) along with low inference latency (8–12 ms per flow), validating the effectiveness of the proposed approach. Despite promising results, this study has certain limitations. The evaluation is limited to a single dataset, and class imbalance affects the detection performance of rare attack classes such as Heartbleed. Future work will focus on incorporating imbalance handling techniques such as SMOTE and weighted loss functions, evaluating the model on additional datasets such as

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UNSW-NB15 and IoT-23, and performing statistical validation through repeated experiments to improve robustness and generalization.

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